AMD DEVELOPER HACKATHON ACT III · HEALTH AND WELLBEING

Clearsheet

Discharge instructions a patient can follow, in their own language. Every line traced to the source and released by a nurse.

Gemma 3 27B on one AMD Instinct MI300X · vLLM on ROCm · team Innova Ottawa

THE PROBLEM

The discharge summary is written for clinicians. The patient takes it home.

```
Furosemide 40 mg PO BID (INCREASED from 20 mg daily).
Metoprolol succinate 50 mg PO daily. Hold if heart rate below 55.
Potassium chloride 20 mEq PO daily - STOPPED. Do not take.
```

Many patients read English as a second language. A language model rewrites and translates this in seconds, and it can also change a dose, drop a "stop", or swap a phone number. Nobody at the bedside can proofread a sheet in Ukrainian or Arabic.

HOW IT WORKS

The model drafts. Code decides what may reach the patient. The nurse decides the rest.

1 Scrub

names, MRN, health card, DOB, phone masked on the hospital side, before any model call

2 Extract

Gemma lists medicines, follow-ups, warning signs, each with a verbatim source quote

3 Ground

quote must exist in the document; every number must sit inside its quote

4 Risk

anticoagulants, insulin, opioids, stops and dose changes always go to a nurse

5 Explain

plain English plus the patient's language

6 Back-check

literal back-translation; numbers and drug names compared in code

7 Release

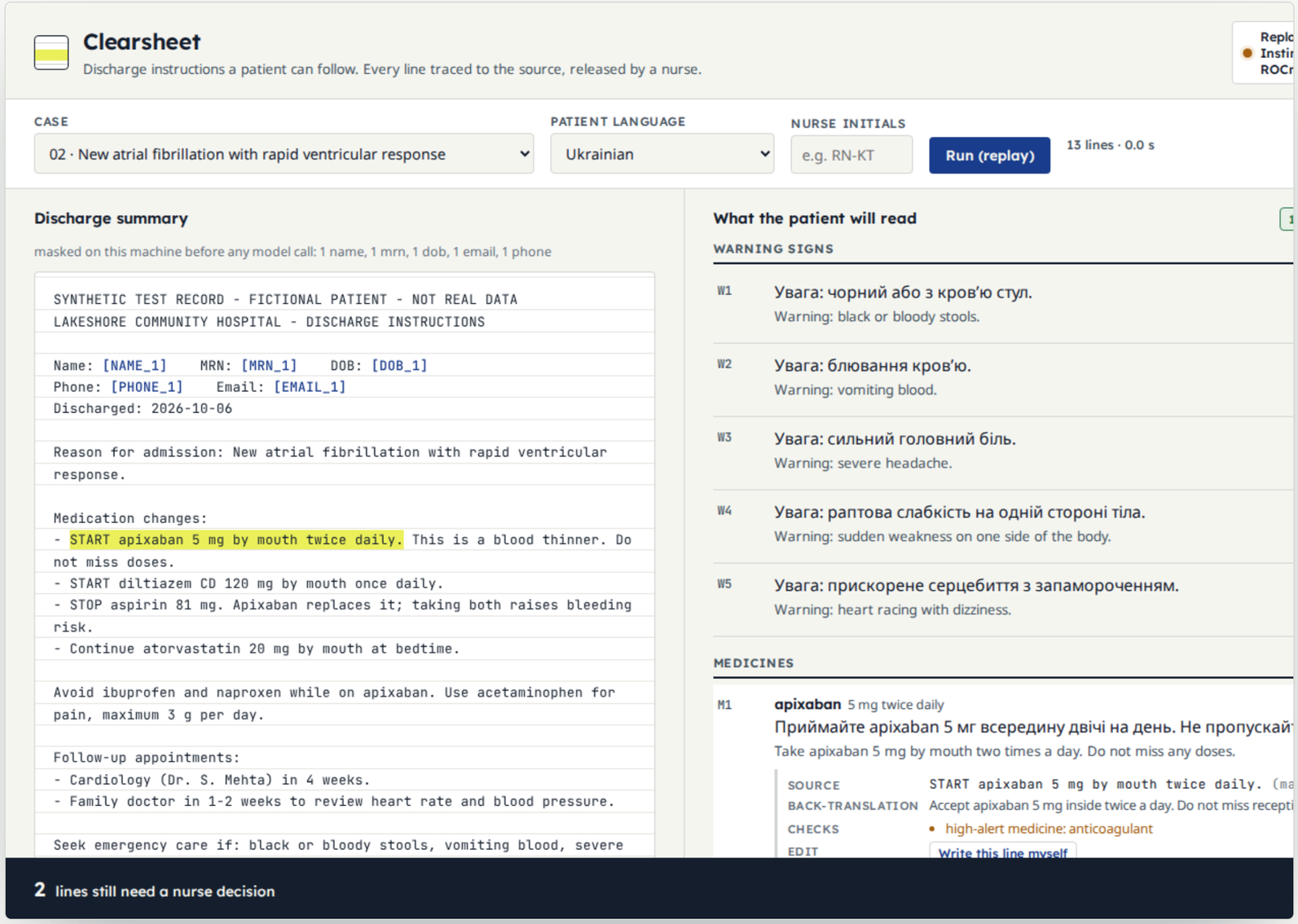
nurse releases flagged lines; blocked lines stay off the sheet

VERIFIED NURSE CHECK BLOCKED

NURSE REVIEW CONSOLE

Click a line, see where it came from.

The highlighter marks the source sentence. The nurse sees the back-translation of what the patient will read.



REAL CATCHES FROM OUR RUNS ON THE MI300X

Gemma translated "call 911" into Ukrainian as "call 103".

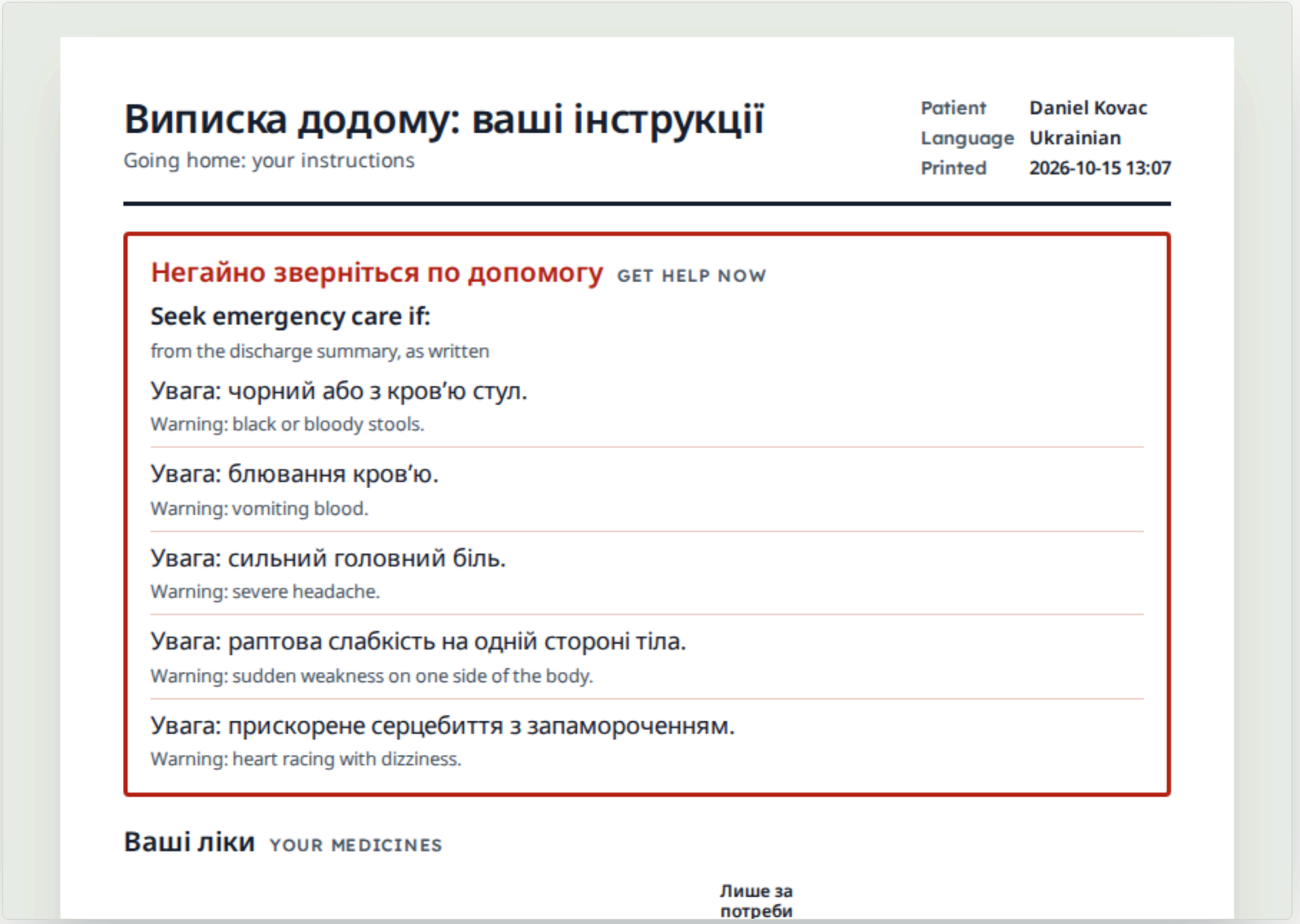
```
EN If you have chest pain, go to the Emergency Department or call 911.
UK … зверніться до відділення невідкладної допомоги або зателефонуйте 103.
```

103 is the ambulance number in Ukraine. The patient is in Canada. The check flags any number that is not in the source, so the line went to the nurse. We then added a prompt rule ("911 stays 911"); the check stays as the safety net.

```
SOURCE Seek emergency care if: black or bloody stools …
DRAFT Tell your doctor if you have black or bloody stools.
→ passed our first checks; the urgency check we added now flags it
```

PATIENT SHEET

Their language first, English underneath, medicines as a pill-box grid.



Виписка додому: ваші інструкції

Going home: your instructions

Patient	Daniel Kovac
Language	Ukrainian
Printed	2026-10-15 13:07

Негайно зверніться по допомогу GET HELP NOW

Seek emergency care if:

from the discharge summary, as written

Увага: чорний або з кров'ю стул.

Warning: black or bloody stools.

Увага: блювання кров'ю.

Warning: vomiting blood.

Увага: сильний головний біль.

Warning: severe headache.

Увага: раптова слабкість на одній стороні тіла.

Warning: sudden weakness on one side of the body.

Увага: прискорене серцебиття з запамороченням.

Warning: heart racing with dizziness.

Ваші ліки YOUR MEDICINES

Лише за потреби

MEASURED ON 8 SYNTHETIC DISCHARGE SUMMARIES × 6 LANGUAGES

What the checks did

97%
medicines found (34 of 35)

100%
of found medicines with the exact dose

94%
new / changed / stopped labelled right

99%
injected wrong numbers and invented quotes blocked (173 of 175)

100%
wrong doses planted in translations flagged (189 of 189)

100%
emergency lines softened to "tell your doctor" flagged (120 of 120)

eval/results.json · 96 model calls · 165,033 tokens · 2026-10-15 13:05 EDT · gemma-3-27b-it · AMD Instinct MI300X (gfx942) · vLLM 0.7.4 V1 engine, ROCm 6.3 · AMD AI Notebooks

AMD INSTINCT MI300X

192 GB on one GPU: a 27B model in bf16 with room to batch a ward's discharges.

PARALLEL REQUESTS	OUTPUT TOK/S	SUMMARIES/MIN (EXTRACT)	P50 LATENCY S
1	40.7	2.9	20.6
8	218.2	15.5	22.3
32	702.6	49.8	27.6
64	1118.0	79.6	35.5

At the batched rate, one translated patient sheet (extract + explain + back-check, about 2390 output tokens) takes about 2.1 GPU-seconds: $0.0012 at $1.99 per MI300X-hour (AMD Developer Cloud list price, 2025).

eval/bench.json · vLLM V1 engine on ROCm · gemma-3-27b-it bf16

WHO PAYS AND WHAT IS NEXT

Hospitals and home-care agencies, per discharge, inside the nurse's existing workflow.

- Runs on the hospital's own AMD GPU or a private endpoint: masked text only, no patient identifiers leave the building.
- It does not diagnose, change doses or add advice. The sheet says so.
- Next: pilot with a newcomer-serving clinic in Ottawa, pharmacist review of the high-alert list, more languages with native-speaker QA.

`github.com/ctmakc/clearsheet · MIT`